HARSH KUMAR BHOJAK

Graduate Researcher · National Institute of Technology (NIT) Jalandhar, Punjab, India

harshbhojak5@gmail.com | harshkb.sp.24@nitj.ac.in | Portfolio Website |
Google Scholar | github.com/harsh2b | linkedin.com/in/harshkb2 | +91-9630499150

RESEARCH INTERESTS

My doctoral research interests converge at the intersection of **Real-Time Interactive AI Systems**, **Virtual Worlds & Serious Games**, and **Voice-Enabled Multimodal Agent Architectures**. Grounded in rigorous graduate foundations of Signal Processing and Machine Learning, my work focuses on:

- **Real-Time Interactive AI & Serious Games:** Building intelligent non-player characters (NPCs) and dynamic behavioral scaffolding inside interactive virtual environments (Unity 3D) via event-driven WebSockets, replacing static scripted game trees with adaptive agency.
- **Voice-Enabled Multimodal Agent Architectures:** Sub-second streaming conversational loops (orchestrating speech recognition, multi-agent LLM reasoning, and low-latency voice synthesis), natural turn-taking dynamics, and dynamic persona modulation for clinical assistive therapy (Autism Spectrum Disorder / ASD).
- **Signal Processing & Applied Machine Learning Foundations:** Mathematical grounding in stochastic processes, statistical signal estimation, temporal state sequences, edge computer vision, and systematic evaluation of multimodal foundation models (Vision-Language Models).

EDUCATION

Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar — Punjab, India
Master of Technology (M.Tech) in Signal Processing and Machine Learning — 2024 – 2026 (Expected)

- **Cumulative GPA:** 8.16 / 10.00 (First Division)
- **Thesis:** *Real-Time Multimodal Voice and Language Interaction in Serious Games for Neurodevelopmental Support*
- **Advisors & Mentors:** Prof. Arun Khosla (Thesis Advisor), Neha Sharma (PhD Scholar & Co-Researcher)
- **Rigorous Graduate Coursework:** Advanced Signal Processing, Applied Linear Algebra and Optimization, Probability and Stochastic Processes, Deep Learning, Statistical Signal Processing, Audio Signal Analysis, Advanced Communication Systems.
- **Research Fellowship:** Awarded MHRD / Ministry of Education Postgraduate Fellowship and Teaching Assistantship.

Indira Gandhi Engineering College (IGEC Sagar / RGPV) — Madhya Pradesh, India
Bachelor of Technology (B.Tech) in Electrical Engineering — 2017 – 2021

- **Cumulative GPA:** 7.50 / 10.00
- **Capstone Project:** *Closed-Loop Standalone 3-Phase Inverter Design and Simulation* (Modeled feedback control systems in PSIM and MATLAB).

PEER-REVIEWED PUBLICATIONS

[1] Harsh Kumar Bhojak, Arun Khosla, and Neha Sharma. **"LLM-Powered Real-Time Voice Agent for Task-Guided Practice in Serious Games for ASD."** In *Proceedings of the IEEE 5th International Conference on Secure Cyber Computing and Communications (ICSCCC)*, IEEE, 2026. [Google Scholar] [IEEE Xplore] [Code Repository]

INDUSTRY & APPLIED AI EXPERIENCE (CONTRACT)

Innodata Inc. — Remote (Contract)
Senior Associate – Vision-Language Models (VLMs) — Jun. 2026 – Present

- Architect complex multi-step reasoning benchmarks and systematic evaluation protocols for frontier Vision-Language Models (VLMs).
- Curate high-fidelity multimodal datasets assessing cross-modal grounding, spatial-geometric relationships, and logical deduction.
- Collaborate on automated rubric verification and synthetic reasoning trajectories to stress-test compositional generalization in frontier foundation models.

Quality Kiosk Technologies Remote (Contract)

AI Research Consultant – Generative AI Sep. 2025 – Dec. 2025

- Standardized 20+ automated evaluation metrics to quantify hallucination rates, semantic drift, and instruction fidelity in generative models, yielding a 30% measured gain in pipeline consistency.
- Executed adversarial red-teaming and safety audits adhering to the OWASP GenAI Top 10 framework, establishing guardrails against prompt injection and data leakage.
- Engineered high-throughput, low-latency API wrappers using FastAPI and Django REST Framework for production inference.

SELECTED RESEARCH SYSTEMS & SOFTWARE

Agentic Voice Scaffolding Engine for Serious Games in ASD Therapy Python, AutoGen, Unity, WebSockets, GCP

- Designed an event-driven `ASR → LLM → TTS` pipeline integrated via WebSockets into Unity serious games to deliver real-time vocal scaffolding for children with Autism Spectrum Disorder (ASD).
- Implemented dynamic persona modulation that adapts instructional pacing to telemetry without modifying game mechanics, demonstrating a 30% increase in patient task completion rates in preliminary user trials.

Wellmate: Clinical Retrieval-Augmented Generation (RAG) Architecture FastAPI, Pinecone, LangChain, Google Cloud Run

- Architected a medical question-answering system combining dense vector search with automated PII masking and clinical hallucination guardrails, achieving >96% contextual relevancy across 1,200+ complex queries.
- Deployed containerized microservices on Google Cloud Run serving 500+ active test users with strict latency SLAs.

Edge-Optimized Urban Traffic & Safety Computer Vision Pipeline PyTorch, YOLOv8, OpenCV

- Engineered an automated urban safety detection pipeline identifying helmet compliance and rider bounding boxes on continuous live video streams.
- Optimized model inference latency for resource-constrained edge computing environments without degradation of mAP.

HONORS & FELLOWSHIPS

- **GATE 2024 Qualified:** Qualified the highly competitive Graduate Aptitude Test in Engineering (GATE 2024), administered nationally by the Indian Institutes of Technology (IITs).
- **MHRD / Ministry of Education Postgraduate Fellowship:** Awarded full government stipend and teaching assistantship for full-time M.Tech studies at NIT Jalandhar (2024 – 2026).
- **First Author IEEE Publication:** Research paper on conversational voice agents in serious games accepted at IEEE ICSCCC 2026.

TECHNICAL SKILLS & METHODOLOGIES

Mathematical Foundations:	Linear Algebra, Optimization, Probability & Stochastic Processes, Digital Signal Processing, Statistical Signal Processing, Audio Feature Analysis (STFT, Mel-Spectrograms, MFCCs).
AI, Machine Learning & Multimodal:	Vision-Language Models (VLMs), Speech Recognition (ASR), Text-to-Speech (TTS), Multi-Agent Systems (AutoGen, CrewAI), RAG, Transformer Architectures, Deep Learning, Computer Vision.
Programming Languages & Frameworks:	Python, MATLAB, JavaScript/TypeScript, PyTorch, TensorFlow, OpenCV, FastAPI, Django REST, LangChain, Hugging Face.
Cloud, Systems & Environments:	Docker, Linux/Bash, Git, Google Cloud Platform (GCP), Pinecone, Unity 3D, LaTeX, Ollama, CI/CD pipelines.

ACADEMIC REFERENCES

Prof. Arun Khosla
Professor, Department of Electronics & Communication Engineering
Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar
Role: Thesis Advisor · `khoslaa@nitj.ac.in`

Neha Sharma
PhD Scholar, Department of Electronics & Communication Engineering
Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar
Role: Research Collaborator & Thesis Co-Mentor